

MCM3AP-Related Peripheral Neuropathy

— Comprehensive Disease Report

Disease Name: MCM3AP-Related Peripheral Neuropathy **MONDO ID:** MONDO:0029131 **OMIM phenotype:** #618124 (Peripheral neuropathy, autosomal recessive, with or without impaired intellectual development, PNRIID) **Gene:** *MCM3AP* (OMIM *603294; HGNC:6946), encoding GANP, chromosome 21q22.3 **Category:** Mendelian, autosomal recessive

Summary

MCM3AP-related peripheral neuropathy is an **ultra-rare, autosomal-recessive, childhood-onset Charcot-Marie-Tooth (CMT) disorder** caused by biallelic (compound heterozygous or homozygous) loss-of-function or hypomorphic variants in *MCM3AP*. This gene encodes **GANP** (Germinal-center-Associated Nuclear Protein), the Sac3-domain scaffold subunit of the human **TREX-2 (transcription-export-2) complex** that docks at the nuclear pore and hands newly transcribed messenger ribonucleoprotein particles (mRNPs) to the NXF1-NXT1 export receptor. The landmark disease-defining study identified biallelic *MCM3AP* variants in nine affected individuals from five unrelated families, presenting with severe childhood-onset primarily **axonal** (four families) or **demyelinating** (one family) CMT, with mild-to-moderate intellectual disability in seven of nine individuals ([PMID: 28633435](#)).

Mechanistically, pathogenic variants deplete GANP, impairing TREX-2-mediated mRNP remodeling and nuclear mRNA export. GANP uses a conserved "trigger loop" to release the DEAD-box helicase DDX39B/UAP56 and complete the mRNP remodeling cycle before export; loss of this function is proposed to drive length-dependent axonal degeneration of peripheral motor and sensory neurons, with a downstream branch involving **R-loop formation and transcription-coupled genome instability**. The mRNA-export defect alters gene expression in a manner **dependent on transcript intron content**, which may explain the selective vulnerability of long-axoned neurons.

The phenotype is a **spectrum** rather than a single presentation. Beyond core sensorimotor CMT with variable cognitive impairment, reported expansions include co-occurring multiple-sclerosis-like CNS demyelination, a motor-neuronopathy/anterior-horn presentation with

widespread fasciculations, ptosis, and (in the earliest human report predating the neuropathy papers) a multisystem picture of immunodeficiency, genomic instability, skin changes, and myelodysplasia — reflecting GANP's additional roles in DNA repair, homologous recombination, and B-cell affinity maturation. A **genotype–phenotype correlation** exists: variants located **outside the Sac3 domain** produce more severe, earlier-onset disease with motor and cognitive impairment than variants within the Sac3 domain. There is currently **no disease-modifying therapy**; management is supportive and rehabilitative.

Section 1 — Disease Information

Overview. MCM3AP-related peripheral neuropathy is a Mendelian, recessively inherited, childhood-onset Charcot–Marie–Tooth (CMT) neuropathy, frequently accompanied by intellectual disability. It belongs to the broad group of inherited peripheral neuropathies but is distinguished by its causal gene (*MCM3AP*/GANP), its role in nuclear mRNA export, and its expanding multisystem spectrum.

Key identifiers.

Resource	Identifier
MONDO	MONDO:0029131
OMIM (phenotype)	#618124
Gene	<i>MCM3AP</i> (OMIM *603294; HGNC:6946); Ensembl ENSG00000160294
Cytogenetic location	21q22.3 (chr21:46,235,133–46,286,297, GRCh38)
Protein	GANP / MCM3AP (UniProt O60318)
MeSH-related	Charcot-Marie-Tooth Disease

Synonyms / alternative names. MCM3AP-related CMT; MCM3AP-associated peripheral neuropathy; recessive Charcot–Marie–Tooth neuropathy with intellectual disability; GANP-related neuropathy; "Peripheral neuropathy, autosomal recessive, with or without impaired intellectual development" (PNRIID). GANP = Germinal-center–Associated Nuclear Protein; *MCM3AP* = Minichromosome Maintenance Complex Component 3 Associated Protein.

Information source. Information is derived from **aggregated disease-level and individual patient reports** in the published literature (case series and small cohorts), not from EHR-scale datasets. Cohorts to date include ~9 individuals in the founding study, a 28-individual retrospective series, and additional single-family reports.

Section 2 — Etiology

Disease causal factors. The disease is **monogenic and genetic**: biallelic pathogenic variants in *MCM3AP*. There is **no environmental or infectious cause**. Inheritance is **autosomal recessive**; affected individuals carry two damaging alleles (homozygous in consanguineous families, or compound heterozygous). GANP is essential — complete null of both alleles is presumed embryonic-lethal (the human disease results from hypomorphic combinations that retain residual function), consistent with animal data showing early requirement.

Genetic risk factors. The sole established genetic risk factor is possession of two pathogenic/hypomorphic *MCM3AP* alleles. Constraint data from gnomAD (see Section 4) show the gene **tolerates heterozygous loss of function** ($pLI \approx 0$), so single-allele carriers are unaffected — fully in keeping with a recessive mechanism.

Environmental risk factors. None identified. Family history / consanguinity increases the a-priori risk of an affected child in the usual autosomal-recessive manner. Age and sex are not established risk modifiers.

Protective factors. No genetic or environmental protective factors have been described specifically for this disease.

Gene–environment interactions. None documented for *MCM3AP*-related neuropathy. The disease is essentially fully genetically determined given a biallelic pathogenic genotype, though **variant-specific residual GANP function** (allelic severity) is the dominant modifier of expression (see Section 4).

Section 3 — Phenotypes

Phenotypes are drawn primarily from the founding cohort ([PMID: 28633435](#)), the 28-individual retrospective series ([PMID: 39228414](#)), and single-family expansions.

Phenotype	Type	Onset	Severity / Course	Frequency	Suggested HPO
Peripheral sensorimotor neuropathy (axonal or demyelinating CMT)	Clinical sign	Childhood	Severe, progressive	Core feature (essentially all)	HP:0009830 (Peripheral neuropathy); HP:0007002 (Motor axonal neuropathy)
Distal muscle weakness / motor developmental delay	Physical manifestation	Childhood	Variable–severe, progressive	Common; delayed independent walking in null genotypes	HP:0002460 (Distal muscle weakness); HP:0001270 (Motor delay)
Intellectual disability / cognitive impairment	Behavioral / cognitive	Childhood	Mild–moderate; "with or without"	7/9 in founding cohort; 46.7% (Sac3) vs 100% (non-Sac3)	HP:0001249 (Intellectual disability)
Areflexia / hyporeflexia	Clinical sign	Childhood	Progressive	Common	HP:0001284 (Areflexia)
Distal sensory loss	Symptom	Childhood	Progressive	Common in classic sensorimotor form	HP:0008936 (Sensory neuropathy)
Widespread fasciculations (motor-neuronopathy variant)	Clinical sign	Variable	Novel, motor-predominant	Rare (single report)	HP:0002380 (Fasciculations)
Ptosis	Physical manifestation	Variable	Mild	Subset	HP:0000508 (Ptosis)
Pes cavus / foot deformity	Physical manifestation	Childhood	Progressive		HP:0001761 (Pes cavus)

Phenotype	Type	Onset	Severity / Course	Frequency	Suggested HPO
				Common in CMT generally	
CNS demyelination (MS-like)	Clinical sign	Adult	Inflammatory	Rare (one family)	HP:0007305 (CNS demyelination)
Immunodeficiency / myelodysplasia / skin changes	Lab + clinical	Variable	Multisystem	Rare (earliest human report)	HP:0002721 (Immunodeficiency); HP:0002863 (Myelodysplasia)

Onset and severity generalities. Onset is typically **childhood** with a **progressive** course. Severity is **variable** and correlates with genotype: null / non-Sac3-domain variants trend toward earlier onset, delayed independent walking, greater motor and cognitive involvement, and higher risk of loss of ambulation.

Quality-of-life impact. No formal EQ-5D / SF-36 / PROMIS data exist for this ultra-rare disease. By analogy to severe childhood CMT, the combination of progressive distal weakness, sensory loss, foot deformity, and (frequently) intellectual disability substantially affects mobility, independence, education, and daily functioning; loss of ambulation is reported in more severe (null) genotypes.

Supporting quotes:

"The variants were associated with severe childhood onset primarily axonal (four families) or demyelinating (one family) Charcot-Marie-Tooth neuropathy. Mild to moderate intellectual disability was present in seven of nine affected individuals." — [PMID: 28633435](#)

"Neurological examination revealed generalized areflexia and widespread fasciculations without sensory abnormalities." — [PMID: 41819534](#)

Section 4 — Genetic / Molecular Information

Causal gene. *MCM3AP* (HGNC:6946; OMIM gene *603294), encoding **GANP**, located at **chromosome 21q22.3** (chr21:46,235,133–46,286,297, GRCh38). The mouse ortholog *Mcm3ap* is on chromosome 10 ([PMID: 27235683](#)).

Protein architecture. GANP is a large, multidomain, bifunctional nucleoporin-associated protein containing: - an N-terminal **HAT-like / FG (phenylalanine-glycine) region** and **histone-acetyltransferase (HAT) domain** ([PMID: 23652018](#)); - **MCM3-acetylating / DNA-replication-licensing** activity; - a C-terminal **Sac3 domain** (the "MCM3AP domain"), homologous to *Saccharomyces* Sac3, which defines the TREX-2 mRNA-export scaffold ([PMID: 27235683](#)).

"GANP, encoded by human chromosome 21, as well as its mouse equivalent on chromosome 10, contains a region homologous to *Saccharomyces* Sac3 that was characterized as a component of the transcription/export 2 (TREX-2) complex and was predicted to be involved in RNA export and metabolism in mammalian cells." — [PMID: 27235683](#)

Pathogenic variants — types and classification. Reported variant classes span **missense, nonsense, frameshift, splice-site, and large multi-exon deletions**; genotypes are typically compound heterozygous or homozygous. Selected reported variants:

Variant	Type	Notes	Source
c.1_5426del (loss of exons 1–25)	Large deletion / likely null	Paternal allele, index Chinese family	PMID: 39228414
c.1858+3A>G	Splice (deletes exon 5)	Maternal allele, index Chinese family	PMID: 39228414
c.5634-1G>T	Splice-acceptor / NMD	Novel splice variant	PMID: 32319184
c.2633G>A (p.Arg878His)	Missense	Recurrent known variant	PMID: 32319184
p.Ile954Thr	Missense (Sac3 domain)	Homozygous, CMT + MS family	PMID: 32954258

Variants are classified **pathogenic** / **likely pathogenic** per ACMG/AMP criteria (null variants meeting PVS1; recurrent missense with functional and segregation support). Additional novel variants have been reported in Lebanese families ([PMID: 29982295](#)) and neuromuscular cohorts ([PMID: 34602496](#)).

Population allele frequency and constraint. gnomAD v2/v4 constraint metrics for *MCM3AP* (ENSG00000160294): **pLI = 3.0e-28 (~0)**, observed/expected LoF (oe_lof) = **0.68** (126 observed vs 184.1 expected LoF; 90% CI/LOEUF 0.59–0.79), missense Z = **−0.14** (no missense constraint), LoF Z = 3.63. These confirm that the gene **tolerates heterozygous loss of function**, consistent with the recessive mechanism — single carriers are unaffected, and disease requires biallelic hits.

Somatic vs germline. All disease-causing variants are **germline**. (Note: the *MCM3AP-AS1* antisense lncRNA is separately implicated in various cancers, but this is unrelated to the neuropathy phenotype and involves somatic/expression dysregulation, not the coding neuropathy variants.)

Functional consequences. Predominantly **loss of function** / **hypomorphic**. Patient fibroblasts from one family showed **severe depletion of GANP** ([PMID: 28633435](#)). Functional work confirmed that different variants perturb GANP differently and that GANP loss alters gene expression **depending on intron content** ([PMID: 32202298](#)).

Modifier genes. No independent modifier genes are established; the principal modifier of severity is the **specific allele combination** and, notably, **variant location relative to the Sac3 domain** (see below).

Genotype–phenotype correlation. In the 28-individual retrospective series ([PMID: 39228414](#)):

"all individuals (100%) with mutations outside the Sac3 domain exhibited early-onset symptoms, motor developmental delays, and cognitive abnormalities, conversely, the proportions of individuals carrying mutations within the domain were 26.7% (motor delays) and 46.7% (cognitive abnormalities)."

Feature	Variants OUTSIDE Sac3 domain	Variants WITHIN Sac3 domain
Early-onset symptoms	100%	Lower
Motor developmental delay	100%	26.7%
Cognitive abnormalities	100%	46.7%

Epigenetic information / chromosomal abnormalities. No recurrent large-scale chromosomal abnormalities (aneuploidy, translocation) cause the disease, though multi-exon deletions occur. GANP itself participates in chromatin modification (its HAT domain modifies chromatin at rearranged immunoglobulin loci), but disease-specific epigenetic signatures are not established.

Section 5 — Environmental Information

MCM3AP-related peripheral neuropathy is a **purely genetic Mendelian disorder**. There are: - **No environmental factors** (toxins, radiation, pollution, occupational exposure) known to cause or trigger it. - **No lifestyle factors** (smoking, diet, exercise, alcohol) established as contributory. - **No infectious agents** implicated.

(Environmental toxins such as vincristine cause peripheral neuropathy through unrelated mechanisms; this is not relevant to the MCM3AP genetic etiology.)

Section 6 — Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic hypomorphic/LoF MCM3AP variants** (missense, splice, frameshift, nonsense, multi-exon deletions) **lead to** reduced or dysfunctional GANP protein (severe GANP depletion demonstrated in patient fibroblasts). (*Demonstrated.*)
2. GANP depletion **results in** loss of the Sac3-scaffold subunit of the nuclear-pore-docked **TREX-2 complex** (GANP–PCID2–DSS1–ENY2–centrin). (*Demonstrated for GANP's TREX-2 role.*)
3. Loss of functional TREX-2 **impairs** handoff of mRNPs to the NXF1–NXT1 export receptor and **impairs** the conserved "trigger-loop"-mediated release of the DEAD-box helicase **DDX39B/UAP56** that completes mRNP remodeling. (*Demonstrated biochemically/structurally.*)
4. Impaired mRNP remodeling **results in** defective **nuclear mRNA export** — the proposed core pathogenic mechanism. (*Demonstrated for mechanism; inferred as disease driver.*)

5. Defective export **alters gene expression in a transcript-intron-content-dependent manner, leading to** selective perturbation of a subset of neuronal transcripts. *(Demonstrated in cell models.)*
6. **Branch A (primary, neuronal):** altered expression of critical transcripts in long-axoned peripheral neurons **leads to length-dependent axonal degeneration** → clinical CMT (motor + sensory). *(Inferred.)*
7. **Branch B (genome instability):** retained/unexported mRNA **leads to R-loop formation and transcription-coupled DNA damage** → contributes to cellular dysfunction and, in some genotypes, multisystem features (immune, hematologic). *(Demonstrated as a consequence of impaired export; contribution to neuropathy inferred.)*
8. Axonal degeneration **results in** distal weakness, sensory loss, areflexia, foot deformity; concurrent effects on CNS neurons **result in** intellectual disability in a genotype-dependent majority. *(Clinical correlation.)*

MCM3AP biallelic variants

↓
 GANP depletion / dysfunction — (also loses HAT / MCM3-acetylation / AID-shepherding roles)

↓
 TREX-2 scaffold loss at nuclear pore

↓
 Failure to release DDX39B/UAP56 → impaired mRNP remodeling

↓
 Defective nuclear mRNA export

↓
 Intron-content-dependent
gene-expression changes

↓
 Length-dependent axonal
degeneration (motor+sensory)

↓
 CMT neuropathy ± intellectual disability

↓
 R-loop formation /
transcription-coupled
DNA damage (genome instability)

↓
 Multisystem features:
immunodeficiency, myelodysplasia,
skin changes (subset)

Detail by category

- **Molecular pathways.** Nuclear mRNA export / TREX-2 pathway (GO:0006406 mRNA export from nucleus; GO:0016973 poly(A)+ mRNA export). NXF1–NXT1 receptor engagement; DDX39B/UAP56 ATPase cycle. GANP also intersects **DNA-damage-induced NF-κB signaling** and **homologous-recombination DNA repair** (via DNA-PKcs).
- **Cellular processes.** Nucleocytoplasmic transport; mRNP assembly/remodeling; DNA double-strand-break repair pathway choice (HR vs NHEJ); chromatin modification (HAT domain); transcription-coupled genome maintenance; neuronal (axonal) maintenance.
- **Protein dysfunction.** Predominantly **loss of function** through protein depletion (null/hypomorphic alleles) or domain-specific impairment (Sac3-domain missense). Variant-specific effects confirmed.
- **Immune-system involvement.** GANP shepherds **activation-induced cytidine deaminase (AID)** to the nucleus and immunoglobulin variable loci, biasing DNA repair toward homologous recombination during somatic hypermutation and antibody affinity maturation — explaining immunodeficiency in the multisystem subset.
- **Tissue-damage mechanisms.** R-loop-driven transcription-coupled DNA damage; genome instability. Length-dependent axonal degeneration in peripheral nerve.
- **Metabolic / metabolomic / lipidomic / proteomic changes.** No disease-specific signatures established.
- **Molecular profiling / transcriptomics.** RNA-seq of GANP/TREX-2-perturbed cells shows altered nucleocytoplasmic ratios of transcript subsets, with intron content and GC content influencing which mRNAs are affected ([PMID: 32202298](#); related structural/RNA-seq work [PMID: 40595470](#)).

Key supporting quotes:

"Accordingly, fibroblasts of affected individuals from one family demonstrated severe depletion of GANP. GANP has been described to function as an mRNA export factor, and to suppress TDP-43-mediated motor neuron degeneration in flies. Thus our results suggest defective mRNA export from nucleus as a potential pathogenic mechanism of axonal degeneration in these patients." — [PMID: 28633435](#)

"Here, we identify the conserved TREX-2 complex as the long-sought factor that facilitates DDX39B/Sub2 to complete the mRNP remodeling cycle." — [PMID: 39862860](#)

"Impaired mRNA export is closely related to DNA damage through R-loop formation." —
[PMID: 39769375](#)

Suggested GO / CL terms. GO:0006406 (mRNA export from nucleus); GO:0016973 (poly(A)+ mRNA export from nucleus); GO:0000724 (double-strand break repair via homologous recombination); GO:0016573 (histone acetylation); GO:0031965 (nuclear membrane); GO:0044613 (nuclear pore central transport channel). Cell types: CL:0000101 (sensory neuron), CL:0000100 (motor neuron), CL:0002573 (Schwann cell), CL:0000540 (neuron).

Section 7 — Anatomical Structures Affected

Organ / system level. - **Primary:** peripheral nervous system (UBERON:0000010) — peripheral nerves (UBERON:0001021), especially long motor and sensory axons. - **Secondary/associated:** central nervous system (UBERON:0001017) — cerebral involvement underlying intellectual disability; CNS white matter in MS-like cases. Anterior horn / spinal cord (UBERON:0002240) in the motor-neuronopathy variant. Skeletal muscle (UBERON:0001134) — neurogenic changes and selective fatty infiltration on MRI. - **Body systems:** nervous (primary); in the multisystem subset also immune/hematopoietic and integumentary (skin).

Tissue / cell level. - Nervous tissue; **peripheral motor and sensory neurons** (axonal degeneration), **Schwann cells / myelin** in demyelinating cases, **anterior horn (lower motor) neurons** in the motor-neuronopathy variant. - Suggested CL terms: CL:0000100 (motor neuron), CL:0000101 (sensory neuron), CL:0002573 (Schwann cell), CL:0000236 (B cell — for AID-related immune features).

Subcellular level. - **Nucleus** and **nuclear pore complex / nuclear envelope** (where TREX-2/GANP operates) — GO:0005643 (nuclear pore), GO:0031965 (nuclear membrane), GO:0005634 (nucleus). Nucleocytoplasmic mRNA transport machinery.

Localization / lateralization. Peripheral neuropathy is **bilateral and symmetric, length-dependent** (distal-predominant, legs before arms) in classic CMT; the reported muscle-MRI pattern in the motor-neuronopathy variant is described as **selective and non-length-dependent** fatty infiltration.

Section 8 — Temporal Development

- **Onset.** Typically **childhood-onset**; described as "severe childhood onset" in the founding cohort. Onset is **insidious/chronic**, not acute. Null / non-Sac3 genotypes present earliest (delayed independent walking, early motor developmental delay).
- **Progression. Progressive** course. Motor developmental delay and, in severe genotypes, eventual **loss of ambulation**. Rate is variable and genotype-dependent (rapid/severe with null alleles; milder/slower with Sac3-domain missense). Electrophysiology in the motor-neuronopathy variant showed a "diffuse mixed acute-on-chronic denervation process."
- **Disease course pattern.** Chronic, progressive, lifelong. The rare co-occurring MS-like CNS demyelination may add a relapsing inflammatory component.
- **Remission / critical periods.** No spontaneous remission. Childhood is the critical window of neurodevelopmental and motor-skill acquisition; any future disease-modifying intervention would ideally act early.

"Electroneuromyography demonstrated diffuse mixed acute-on-chronic denervation process. Whole-body muscle MRI showed a selective non-length-dependent pattern of fatty infiltration." — [PMID: 41819534](#)

Section 9 — Inheritance and Population

Epidemiology. Ultra-rare. No formal prevalence or incidence estimates exist (fewer than ~40–60 individuals reported worldwide across all cohorts). Not listed with a specific population-scale prevalence figure in Orphanet.

Inheritance and genetics. - **Inheritance pattern: Autosomal recessive** (compound heterozygous or homozygous). Confirmed by segregation and by gnomAD constraint showing heterozygous-LoF tolerance. - **Penetrance:** Appears high/complete for the neuropathy phenotype in biallelic individuals, though **expressivity is highly variable** (severity and cognitive involvement genotype-dependent). - **Expressivity: Variable** — from milder Sac3-domain phenotypes to severe early-onset non-Sac3 phenotypes, and multisystem presentations. - **Genetic anticipation:** Not applicable (not a repeat-expansion disorder). - **Germline mosaicism:** Not reported. - **Founder effects:** No established founder mutation; recurrent variants (e.g., p.Arg878His) noted across reports. - **Consanguinity:** Plays the

expected role in homozygous cases (e.g., consanguineous families with homozygous Sac3-domain missense). - **Carrier frequency:** Not precisely established; gnomAD LoF observed count (126) implies rare heterozygous carriers in the general population; biallelic disease is correspondingly very rare.

Population demographics. Reported across multiple ethnicities/geographies (European, Chinese, Lebanese, others) — **no single population enrichment**. **Sex ratio** is not reported to be skewed (autosomal recessive; expected ~1:1 male:female). Age distribution centers on **pediatric onset** with lifelong course.

Section 10 — Diagnostics

Clinical / electrophysiology. - **Nerve conduction studies / EMG:** define CMT subtype — reduced amplitudes (axonal) or slowed conduction velocities (demyelinating). In the motor-neuronopathy variant, ENMG showed a "diffuse mixed acute-on-chronic denervation process." - **Whole-body / muscle MRI:** the motor-neuronopathy case showed a "selective non-length-dependent pattern of fatty infiltration," a potentially distinguishing imaging clue. - **Neurological exam:** distal weakness, sensory loss, areflexia/hyporeflexia, pes cavus; in the motor variant, generalized areflexia with widespread fasciculations and no sensory abnormality. - **Nerve biopsy:** not routinely required; would show axonal loss (axonal form) or demyelination/onion-bulbs (demyelinating form) — nonspecific.

Genetic testing (definitive). - **Recommended approach:** since phenotype overlaps many CMT genes, **next-generation sequencing (WES/WGS or a large inherited-neuropathy gene panel including MCM3AP)** is the diagnostic route. Single-gene testing is reasonable only when phenotype and family history are highly suggestive. - **WES/WGS:** high utility — the disease gene was discovered by exome sequencing across multiple centers via gene-matching. WGS/CNV analysis is important because **large multi-exon deletions** and **splice variants** occur and may be missed by panel/coverage gaps. - **Chromosomal microarray / CNV analysis:** useful to detect multi-exon deletions (e.g., c.1_5426del). - **Confirmatory functional testing:** GANP protein quantification (Western blot) in patient fibroblasts can support pathogenicity (severe depletion demonstrated). - **Not applicable:** karyotyping, FISH, mtDNA testing, and repeat-expansion testing are not indicated.

Omics-based diagnostics. RNA-seq can demonstrate splice consequences (e.g., NMD-mediated transcript loss) and intron-content-dependent expression changes, useful for variant interpretation. No proteomic/metabolomic clinical assay exists.

Clinical criteria / differential diagnosis. No formal consensus diagnostic criteria beyond CMT classification plus molecular confirmation. Differential diagnosis includes other **recessive childhood-onset CMT genes** (e.g., *SH3TC2*, *GDAP1*, *IGHMBP2*), CMT-with-intellectual-disability syndromes, hereditary motor neuronopathies / SMA (for the motor-neuronopathy variant), and — in the MS-overlap family — acquired inflammatory CNS demyelination. Molecular genetic testing distinguishes MCM3AP disease.

Screening. For at-risk families: **cascade / carrier testing** of relatives and **prenatal / preimplantation genetic testing** once the familial variants are known. No population newborn screening exists.

Section 11 — Outcome / Prognosis

- **Survival / mortality.** Not a directly life-limiting neuropathy in most reports; **no specific survival or mortality statistics** are available. The multisystem/immunodeficiency-myelodysplasia subset could carry additional risk, but data are limited to single reports.
- **Morbidity / function.** Substantial. Progressive distal weakness and sensory loss cause mobility impairment; severe (null) genotypes may lose ambulation. Intellectual disability (mild–moderate) adds educational and functional burden in the majority. Formal disability/QoL measures have not been applied.
- **Complications.** Foot deformities, contractures, falls, and complications of immobility; in demyelinating-CNS and immunodeficiency subsets, additional neurological or infectious complications.
- **Recovery potential.** No spontaneous recovery; the process is progressive and there is no curative therapy.
- **Prognostic factors.** The strongest predictor is **genotype: null / non-Sac3-domain variants** predict earlier onset, delayed walking, greater motor and cognitive impairment, and higher risk of loss of ambulation; **Sac3-domain missense** predicts milder disease. No molecular prognostic biomarker beyond genotype is established.

"affected individuals with null mutations presented with delayed independent walking" — [PMID: 32319184](#)

Section 12 — Treatment

There is currently **no disease-modifying or curative therapy**. Management is **supportive, symptomatic, and rehabilitative**, mirroring general CMT care.

- **Pharmacotherapy.** No targeted drug. Symptomatic agents may include neuropathic-pain medications (e.g., gabapentinoids, duloxetine) where painful sensory symptoms occur (NCIT clinical-intervention category: analgesic / neuropathic-pain pharmacotherapy). No pharmacogenomic guidance specific to *MCM3AP*.
 - **Advanced therapeutics.** None approved. Gene therapy, RNA-based therapies, cell therapy, and targeted molecular therapies are **conceptually plausible future directions** (see follow-up) but **not available**. Because the mechanism is loss of function, gene-replacement / augmentation strategies are theoretically rational.
 - **Surgical / interventional.** Orthopedic correction of foot deformities (e.g., tendon transfer, osteotomy) and management of contractures as needed (NCIT: orthopedic surgical procedure).
 - **Supportive / rehabilitative. Physical therapy, occupational therapy** (NCIT:C15533 Rehabilitation Therapy; NCIT:C15629 Physical Therapy; NCIT:C15236 Occupational Therapy), orthoses (ankle-foot orthoses), assistive/mobility devices, fall prevention; **speech/ educational support** for intellectual disability; multidisciplinary neuromuscular care.
 - **Experimental.** No disease-specific registered clinical trials identified.
 - **Treatment strategy.** Individualized, multidisciplinary, supportive care coordinated by neuromuscular specialists, with genetic counseling for the family.
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Section 13 — Prevention

- **Primary prevention.** Because the disease is monogenic and recessive, "prevention" operates at the **reproductive-genetics** level rather than through lifestyle/environmental modification. Options include **genetic counseling**, **carrier testing** of relatives, and, for known-carrier couples, **preimplantation genetic testing (PGT)** or **prenatal diagnosis**.
- **Secondary prevention.** Early molecular diagnosis enables early rehabilitative intervention, orthotic management, educational support, and surveillance for complications.

- **Tertiary prevention.** Prevent complications of progressive weakness (contractures, falls, skin breakdown) via PT/OT, orthoses, and monitoring; in the multisystem subset, monitor immune/hematologic status.
- **Counseling. Genetic counseling** is central: recurrence risk is 25% for future children of two carrier parents; carrier and cascade testing offered to at-risk relatives.
- **Immunization / public-health / environmental interventions.** Not applicable (no infectious or environmental etiology). Standard immunizations advisable generally, particularly if immunodeficiency features are present.

Section 14 — Other Species / Natural Disease

- **Taxonomy / orthologs.** *MCM3AP*/GANP is **evolutionarily conserved**. Human *MCM3AP* (chr21); mouse *Mcm3ap* (chromosome 10, NCBI Gene ID 54387). Functional orthologs of the Sac3/TREX-2 scaffold exist in *Drosophila* ("Xmas-2" / protein x-mas-2), budding yeast (**Sac3**, in the yeast TREX-2 complex), and other eukaryotes.
- **Natural disease in other species.** No naturally occurring companion-animal or wildlife disease attributable to *MCM3AP* is documented in OMIA; the disease is currently described only in humans.
- **Comparative biology.** The mechanism (TREX-2-mediated mRNA export and mRNP remodeling; DDX39B/Sub2 release via a conserved trigger loop) is **strikingly conserved from yeast to humans**, making invertebrate and yeast systems informative for mechanism even though they do not model the clinical neuropathy.
- **Transmission.** Not applicable (non-infectious, non-zoonotic).

Section 15 — Model Organisms

- ***Drosophila melanogaster*.** GANP (via its ortholog) **suppresses TDP-43-mediated motor-neuron degeneration** in flies — the most direct disease-relevant model linking GANP to motor-neuron protection (cited in [PMID: 28633435](#)). Insect studies (fly/mosquito/beetle) also confirm *MCM3AP*'s essential developmental role (knockdown lethality), supporting essentiality.

- **Mouse (*Mcm3ap*).** Genetically modified mice exist, including conditional *ganp*-knockout and *ganp*-transgenic lines used primarily to study immune/germinal-center biology and tumorigenesis; GANP expression is "highly associated with tumorigenesis in mice and humans," demonstrated "using genetically modified mice" (PMID: 39769375; immune models PMID: 23652018, PMID: 27235683). A dedicated mouse model recapitulating the **peripheral neuropathy** phenotype is **not yet established** — a key gap.
- **Chicken DT40 B-cell line.** Used to dissect GANP's role in DNA-repair pathway choice (HR vs NHEJ) during AID-dependent Ig diversification (PMID: 24808370).
- **Yeast (*Saccharomyces cerevisiae*).** Sac3/TREX-2 provides the mechanistic template for the human complex; ortholog Thp3–Csn12 informs the mRNP surveillance/decay branch.
- **Patient-derived cells.** Fibroblasts (GANP depletion; DNA-repair assays) and cell-line RNA-seq models are the principal human-relevant systems used to date; iPSC-derived neurons/organoids would be a logical next model.

Phenotype recapitulation / limitations. Existing genetic models capture GANP's **molecular** roles (mRNA export, DNA repair, immune maturation, tumorigenesis) well, but **no current model fully recapitulates the human peripheral-neuropathy-with-intellectual-disability phenotype**. The fly TDP-43 model captures motor-neuron protection but not the full clinical spectrum.

Mechanistic Model / Interpretation (synthesis)

MCM3AP-related neuropathy is best understood as a "**mRNP-export-opathy**" of long neurons. GANP sits at the nuclear pore as the Sac3 scaffold of TREX-2, orchestrating the final steps of mRNP maturation: it engages the NXF1–NXT1 export receptor and, via a conserved trigger loop, evicts DDX39B/UAP56 to license export. When biallelic variants deplete or cripple GANP, this handoff fails. The consequence is not global translational collapse but a **selective, intron-content-dependent distortion of the transcriptome** — which plausibly explains why the longest, most transcript-demanding cells (peripheral motor/sensory axons and, less severely, CNS neurons) are preferentially affected, yielding length-dependent axonal degeneration and intellectual disability. A parallel branch — R-loop-driven, transcription-coupled genome instability — provides a unifying explanation for the **multisystem outliers** (immunodeficiency, myelodysplasia, skin changes) reported in the earliest human case and for GANP's established roles in AID shepherding and HR-biased DNA repair.

The **Sac3-domain genotype-phenotype axis** is the report's most clinically actionable synthesis: variants **outside** the Sac3 domain (often more disruptive to the protein overall / closer to null) uniformly cause early-onset motor-and-cognitive disease, whereas Sac3-domain missense variants (which may retain partial scaffold function) cause milder disease. This gradient, together with gnomAD constraint showing heterozygous-LoF tolerance, cements a **loss-of-function, dosage-sensitive recessive model** in which residual GANP activity sets severity.

Evidence Base

PMID	Title (abbrev.)	Contribution
28633435	<i>MCM3AP in recessive CMT and mild intellectual disability</i>	Landmark: establishes MCM3AP/GANP as the disease gene; 9 individuals/5 families; axonal vs demyelinating; ID in 7/9; GANP depletion; mRNA-export mechanism
28969388	<i>Biallelic MCM3AP mutations cause CMT with variable presentation</i>	Second 2017 cohort confirming variable clinical spectrum
39228414	<i>Biallelic MCM3AP mutations (28 individuals)</i>	Genotype–phenotype: Sac3 vs non-Sac3 severity; novel c.1_5426del and c.1858+3A>G
32954258	<i>Recessive CMT and MS with MCM3AP variant</i>	Phenotype expansion: CMT + CNS demyelination; p.Ile954Thr; MCM3-acetylation role
41819534	<i>Motor neuronopathy with fasciculations in MCM3AP disorder</i>	Novel anterior-horn/motor-neuronopathy phenotype; ENMG and muscle-MRI findings
32319184	<i>Novel MCM3AP variants</i>	c.5634-1G>T and p.Arg878His; null genotypes → delayed independent walking
32202298	<i>Distinct effects on GANP; intron-content-dependent expression</i>	Variant-specific GANP effects; intron-content-dependent gene-expression mechanism
26615982	<i>De novo MCM3AP + POMP; immunodeficiency/genomic instability</i>	Earliest human report; GANP DNA-repair/HR/NF-κB/AID roles; multisystem phenotype
27235683	<i>GANP in RNA metabolism for B-cell maturation</i>	Gene location, mouse ortholog, Sac3/TREX-2 domain
23652018	<i>GANP HAT domain modulates AID recruitment</i>	HAT domain / chromatin-modifying function
20507984	<i>GANP-mediated recruitment of AID</i>	AID shepherding — second major GANP function
24808370	<i>GANP regulates DNA repair pathway choice via DNA-PKcs</i>	HR-vs-NHEJ pathway choice; DT40 model

PMID	Title (abbrev.)	Contribution
39769375	<i>Tumorigenesis from aberrant GANP in TREX-2</i>	R-loop/genome-instability branch; mouse models
39862860	<i>Structures and mRNP remodeling of TREX-2</i>	TREX-2 completes DDX39B/Sub2 mRNP remodeling cycle
40595470	<i>DDX39B regulation by TREX-2</i>	Trigger-loop mechanism; GC-content-dependent mRNA effects
39384042	<i>NS1-BP/NXF1 mRNA export</i>	Confirms GANP–NXF1 partnership in export

Evidence-type distribution: human clinical (case series/cohorts) for phenotype, inheritance, and genotype–phenotype; human patient-derived cells for GANP depletion and DNA-repair defects; in vitro/structural biochemistry for the TREX-2/DDX39B mechanism; model-organism (fly, mouse, DT40, yeast) for GANP's molecular and immune functions; computational (gnomAD) for constraint.

Limitations and Knowledge Gaps

1. **Ultra-rare disease, small n.** Total reported individuals number only in the low tens; prevalence, incidence, survival, QoL, and natural-history data are essentially absent.
2. **Mechanism partly inferred.** Defective nuclear mRNA export is **demonstrated at the molecular level and strongly implicated**, but the direct causal link from export defect to peripheral **axonal degeneration** in humans remains **inferred**, not proven in a neuronal disease model.
3. **No neuropathy-specific animal model.** Existing mouse models address immunity and tumorigenesis, not the peripheral-neuropathy phenotype; the fly TDP-43 model is partial.
4. **Genotype–phenotype correlation is retrospective** and based on limited cohorts; the Sac3-domain rule needs prospective validation and functional dissection.
5. **Selective neuronal vulnerability unexplained mechanistically** — why long axons and cognition are targeted while most tissues are spared is not fully resolved (the intron-content hypothesis is promising but incomplete).

6. **Multisystem spectrum boundaries unclear** — the relationship between the neuropathy-dominant presentations and the immunodeficiency/myelodysplasia presentations (allelic series? modifier genes?) is not established.
 7. **No therapeutics** and no biomarkers of progression beyond genotype.
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Proposed Follow-up Experiments / Actions

1. **Build a neuronal disease model.** Generate patient-derived **iPSC motor/sensory neurons and cerebral organoids**, plus a conditional neuron-specific *Mcm3ap* hypomorph mouse, to test whether GANP loss produces axonal degeneration and to define the affected transcript set in neurons.
 2. **Define the vulnerable transcriptome.** Perform nucleocytoplasmic fractionation RNA-seq in patient neurons to map which intron-/GC-rich transcripts are export-impaired, linking specific mis-exported mRNAs to axonal maintenance.
 3. **Functionally validate the Sac3 rule.** Systematically assay GANP protein level, TREX-2 assembly, DDX39B release, and export activity for Sac3 vs non-Sac3 variants to convert the clinical correlation into a mechanistic severity scale usable for variant classification.
 4. **Test the R-loop/genome-instability branch in neurons.** Quantify R-loops and DNA damage in patient neurons; determine whether this branch contributes to neurodegeneration or is confined to the immune/hematologic phenotype.
 5. **Establish a natural-history registry.** Aggregate cases internationally (via GeneMatcher/consortia) to obtain prevalence, progression rates, ambulation outcomes, cognitive trajectories, and validated QoL measures.
 6. **Explore gene-augmentation feasibility.** Given the loss-of-function mechanism and heterozygous-LoF tolerance, evaluate AAV- or mRNA-based GANP restoration in cellular/animal models as a proof-of-concept therapeutic direction.
 7. **Standardize diagnostics.** Ensure inherited-neuropathy panels/WGS pipelines capture large multi-exon deletions and splice variants in *MCM3AP*, and incorporate GANP Western blot / RNA-seq as functional confirmation for VUS resolution.
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Report compiled from 10 confirmed findings and 33 reviewed papers over 5 investigation iterations. Evidence types span human clinical cohorts, patient-derived cells, structural/in-vitro biochemistry, model organisms, and computational constraint analysis. Claims are attributed to primary literature by PMID with verbatim supporting quotes where provided.

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